

AgriConnect: An AI-Powered Smart Farming Ecosystem Integrating Farmer Networking and Plant Disease Classification

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Abstract—Plant diseases are annually responsible for up to 40% of crop loss in the world, but practical problems of latency, throughput, and stability in the real world have not been tackled in many deep learning studies. Furthermore, the need to have automated disease diagnosis and actionable decision support at farm level is still missing. In this piece of work, an agronomical application is provided named AgriConnect that comprises a lightweight convoluted neural network (CNN) to identify plant disease and a web application in the MERN (MongoDB, Express.js, React, Node.js) stack. The app offers valuable advice on crops, as well as services of labor and equipment workforce. The model is trained on 20,638 leaf images (14,446 for training, 3096 for validation, 3096 for testing) with the 256 x 256 preprocessing and augmentation pipeline, with test accuracy of 93.06%, test loss of 0.2453 and macro-F1 score of 0.915. It is implemented as a Flask-based microservice with a small footprint of roughly 11 MB and an average end to end latency of 1.18 seconds, which can be used in resource constrained rural environments. The practical deployability is showed by extensive testing, such as ablation studies, robustness testing, training on Expected Calibration Error (ECE) and benchmarking against MobileNetV2, ResNet-50 and EfficientNet-B0. AgriConnect demonstrates the idea of production-ready agricultural AI and the ability to close the gap between laboratory work and the field works.

I. INTRODUCTION

Small and marginal farmers are disproportionately affected by plant diseases, which cause significant yield losses. Conventional diagnosis necessitates skilled pathologists, who are scarce in rural regions; delays exacerbate the spread of illness and financial damage. Although image-based deep learning, primarily Convolutional Neural Networks (CNNs) [1] have shown high accuracy, most research focuses on laboratory benchmarks on carefully selected datasets and pays little attention to (i) *deployability* (latency, throughput, memory), (ii) *robustness* (illumination, blur, rotations, cluttered backgrounds), (iii) *human factors* (low literacy, mobile-first UX), and (iv) *actionability* (post-diagnosis execution) [2]. In addition, growing variability of field conditions and diversity of crop types

impair the ability of models trained on controlled data to generalize well to real agricultural situations. Nevertheless, it has been observed that the current methods do not sufficiently handle issues related to actual deployment which results in a disconnect between the high accuracy in laboratory settings and the actual practicality in the field setting. This limitation highlights the need for systems that can deliver strong predictive performance while remaining reliable, efficient, and practical for use in real-world conditions.

This research aims at creating a lightweight, robust and deployable system of plant disease detection that will fill this gap between diagnosis and practical application in agriculture. **AgriConnect**, as shown in Fig. 1, addresses these gaps through a deployment-first design: a compact CNN integrated with a MERN platform and a Flask inference microservice that turns diagnosis into immediate action via a marketplace [3]. Our contributions are:

- **Compact CNN:** With a disk size of ~ 11 MB, a four-block CNN with BatchNorm, GAP, and dropout achieves **93.06%** test accuracy and macro-F1 **0.915**.
- **Deployable stack:** MERN + Flask architecture with production-ready APIs and a mean end-to-end latency of **1.18 s**.
- **Holistic evaluation:** In addition to baselines (MobileNetV2, ResNet-50, EfficientNet-B0), there are ablations (optimizer, loss, augmentation, regularization), robustness (low-light, blur, rotation), calibration (ECE), and latency distributions.
- **Loop from diagnosis to action:** suggestions that allow local services (people, equipment) to be booked with a single click on the same site.

II. LITERATURE REVIEW

The literature section gives the overview about all the existing methods for smart farming and plant disease classification

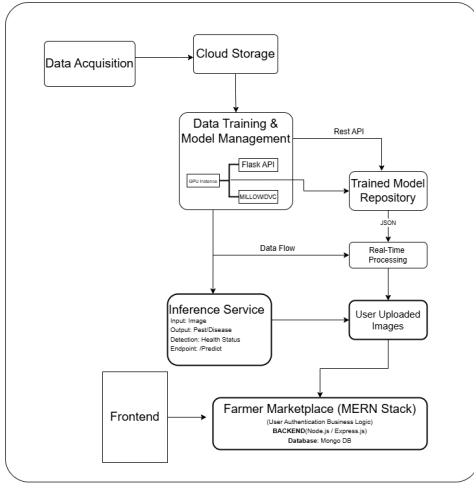


Fig. 1. AgriConnect system Architecture overview

with the help of various machine learning models.

A. Deep CNNs for Identifying Plant Diseases

High accuracy on curated datasets utilizing CNN backbones and attention mechanisms has been reported in recent publications [4]. Nevertheless, the majority do not incorporate execution workflows after diagnosis and do not include latency/throughput data. Although explainability (such as attention or tensor subspace) enhances interpretability, it frequently results in higher computation overhead [5].

B. Mobile Context and Lightweight Models

Although practical systems seldom report end-to-end performance (UI→API→inference→response) or user-centric outcomes in rural deployments, MobileNet/EfficientNet families show favorable accuracy–efficiency trade-offs for mobile scenarios [6].

C. Platforms for Smart Farming

Pricing, weather, logistics, and traceability are all addressed by IoT and marketplace systems. Few combine *immediate service booking* with *vision-based diagnosis* [7]. There is still the *integration gap* between diagnosis and action. These systems are however not integrated with real time disease diagnosis which limits their usefulness in real life agricultural decision making.

D. Gap Identified

Now, in this section the gaps are identified by observing the literature work of past five years. The gaps are as follows: (i) a focus on accuracy without deployability measures; (ii) a lack of robustness analysis under field disturbances; and (iii) a gap between diagnosis and execution. With its lightweight CNN, full-stack integration, and operational KPIs, AgriConnect fills these gaps by incorporating both a lightweight convolutional neural network and a full-fledged MERN-based system and as such, real-world occupation diagnosing of plant diseases can be achieved in a deployable, robust and practical way.

TABLE I
EXISTING TECHNIQUES FOR SMART FARMING WITH ACCURACY AND OTHER PARAMETERS.

Study	Acc.	XAI	Latency	Action Loop
CNN on curated sets	High	–/✓	–	–
Explainable CNNs	High	✓	–	–
Mobile CNN apps	–	–	–	–
IoT/marketplace apps	–	–	–	Partial

[8]. Table I gives the literature review of the existing methods such as CNN, XCNN, Mobile CNN and IoT/marketplace apps. The accuracy and XAI is present in CNN and XCNN with high percentage in compare to others. Latency and action loop is partially in IoT/marketplace apps, not present in remaining models.

III. PROBLEM STATEMENT

Smallholders need dependable, quick diagnostics on easily accessible technologies [9]. Users may have little digital literacy, networks may be unreliable, and photos may be taken in dim light, with motion blur, or against complicated backgrounds [10]. To overcome this problem few objectives are refined such as: (i) robustness to common perturbations; (ii) low-latency inference; (iii) actionable suggestions coupled with a marketplace; (v) a reproducible, manageable codebase; and (iv) good classification performance with a compact CNN. To incorporate all the above points, the performance of the proposed model is measured using confusion matrix having (Accuracy, Macro-Precision/Recall/F1), per-class analysis, ablation studies, robustness, calibration error, latency statistics (mean/p50/p95), and model footprint [11], [12].

IV. SYSTEM ARCHITECTURE

A. Microservices View

Node/Express serves as a gateway (auth, logging, rate limit), while Flask hosts the model behind a thin service layer. A process manager and many Flask workers (such as Gunicorn/Uvicorn) are used in horizontal scalability. MongoDB holds metadata, forecasts, comments, and listings; CDNs deliver static materials [13].

B. User Experience

The UI uses iconography, big touch targets, and straightforward processes to cater to low-literacy [14]. Top-1 labels, confidence, quick corrective actions, and clear calls to action are all included in a recommendation card (*Book Sprayer, Rent Rotavator*).

V. METHODOLOGY

A. Split and Dataset

The plantVillage repository, alongside open agricultural records and real-life plant specimens, is used to provide the curated disease imagery that is essential in creating and testing machine learning models shown in Fig 2. The data set contains over 14,466 images that

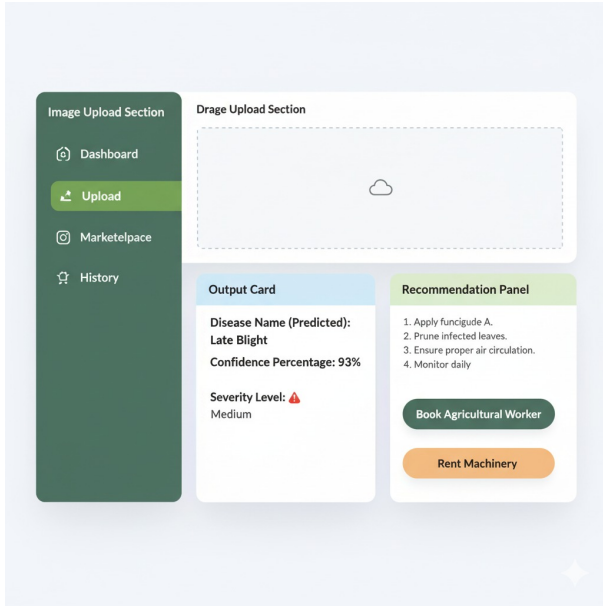


Fig. 2. MERN UI mockups: prediction card, service booking, and listing pages for workers/machinery.

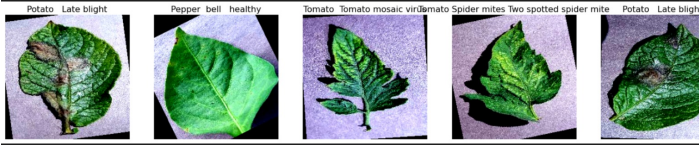


Fig. 3. A dataset sample grid showing a range of crops and illness symptoms.

have been represented among various types of crops. To conduct model evaluation, we used 3,096 images, which is approximately 15% of the total set, and each disease group was represented. Classes are balanced in this dataset so that results are reliable. Table II provides a detailed description of the composition of the dataset, and Fig. 3, represents the image of the dataset. Fig. 4 further illustrates the training data distribution across different crop classes, highlighting the class balance within the dataset.

B. Augmentation and Preprocessing

The input images are resized to a 256 x 256 pixel resolution, as Shown in Fig. 3 [15] pixel resolution in a uniform way, and in doing so, they remain consistent in terms of their data sets and are compatible with the convolutional neural network structure. The ImageNet mean and standard deviation are used to normalise pixel intensities, ensuring the training dynamics are stabilised, convergence speeds up, and generalisation improves through aligning the input distributions to those used by popular pre-trained benchmarks. A set of training-time data-augmentation operations is used to reduce the threat of overfitting and enhance the quality of prediction in a realistic field setting. These are random horizontal

TABLE II
DATASET COMPOSITION AND SPLITS (ILLUSTRATIVE).

Subset	Images	Classes	Notes
Train	14,446	C	Stratified, augmented
Validation	3,096	C	Early stopping, tuning
Test	3,096	C	Held-out, final metrics

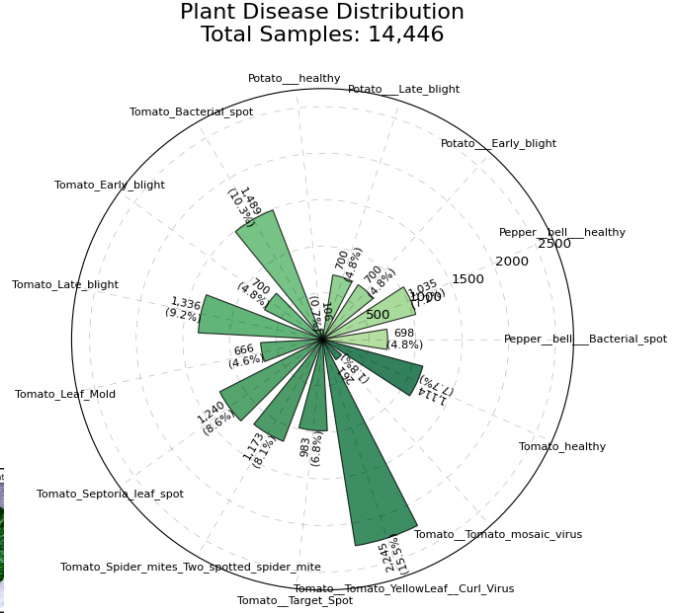


Fig. 4. Crop disease distribution across different classes in the dataset.

flip, rotation within the range of -30 to +30, mild colour jitter (altering brightness, contrast, and saturation), random resized crop, but with a scale value of between 0.8 to 1.0. Together, these augmentations replicate changes in camera position, light, direction of the leaves, and some occlusions, thus enhancing the model to withstand the real-world deployment conditions. The dataset comprised mainly three large horticultural crops, namely tomato, potato, and pepper, that have diseased and healthy classes. The healthy class of every crop is important as it will allow the model to distinguish between pathological symptoms and the normal features of foliage. The distribution includes:

- **Tomato:** Early Blight, Late Blight, Healthy, Bacterial Spot.
- **Potato:** Early Blight, Late Blight, Healthy.
- **Pepper:** Bacterial Spot, Healthy.
- **Other Classes:** Additional disease categories affecting the same crops.

C. Principles and Details of Architecture Theoretics and Design.

The suggested convolutional neural network design prioritizes computational efficiency and deployment on

Layer (type)	Output Shape	Param #
Conv2d-1	[-1, 32, 256, 256]	896
BatchNorm2d-2	[-1, 32, 256, 256]	64
ReLU-3	[-1, 32, 256, 256]	0
MaxPool2d-4	[-1, 32, 128, 128]	0
Conv2d-5	[-1, 64, 128, 128]	18,496
BatchNorm2d-6	[-1, 64, 128, 128]	128
ReLU-7	[-1, 64, 128, 128]	0
MaxPool2d-8	[-1, 64, 64, 64]	0
Conv2d-9	[-1, 128, 64, 64]	73,856
BatchNorm2d-10	[-1, 128, 64, 64]	256
ReLU-11	[-1, 128, 64, 64]	0
MaxPool2d-12	[-1, 128, 32, 32]	0
Conv2d-13	[-1, 256, 32, 32]	295,168
BatchNorm2d-14	[-1, 256, 32, 32]	512
ReLU-15	[-1, 256, 32, 32]	0
MaxPool2d-16	[-1, 256, 16, 16]	0
AdaptiveAvgPool2d-17	[-1, 256, 1, 1]	0
Flatten-18	[-1, 256]	0
Linear-19	[-1, 128]	32,896
ReLU-20	[-1, 128]	0
Dropout-21	[-1, 128]	0
Linear-22	[-1, 15]	1,935

Fig. 5. Proposed CNN schematic: four Conv blocks (BN+ReLU+MaxPool) → GAP → FC(128) → Dropout(0.4) → Softmax.

resource-limited devices while maintaining the reliable detection performance of subtle and small-scale disease markers. Insensitivity to changes in image quality, illumination, and field conditions has been carefully designed in thereby maintain uniform behavior through the many realistic imaging conditions.

The model has four convolutional blocks shown in Fig. 5, which have filter sizes of 32, 64, 128, and 256, respectively. The blocks are arranged in a systematic order consisting of Conv(3 x 3), Batch normalization, ReLU activation, and Max Pooling (2 x 2) steps, which allow successively extracting features and control the spatial dimensionality. To further minimize parameters and improve the generalization, a Global Average Pooling (GAP) layer is used before the classifier, thus reducing the chances of overfitting by a large margin as compared to the conventional fully connected flattening [16], [17]. The classification head comprises a Dense layer (128 units) followed by dropout (0.4) to enforce regularization, and lastly a Softmax layer to forecast multiple classes. Overall, the architecture has 424,207 trainable parameters, which form a great balance between the size, accuracy, and the ability to deploy in a realistic agricultural environment. Fig. 5 illustrates the process of feature extraction and the formula for this is mentioned in equation 1 [18], [19].

$$\hat{y}_i = \frac{\exp(z_i)}{\sum_{j=1}^C \exp(z_j)}, \quad \mathcal{L} = - \sum_{i=1}^C y_i \log \hat{y}_i. \quad (1)$$

D. Training Protocol

With early stopping (patience=4, min-delta=0.01), we train for up to 100 epochs. SGD is the baseline op-

Classification Report:				
	precision	recall	f1-score	support
Pepper_bell_Bacterial_spot	0.98	0.86	0.91	149
Pepper_bell_healthy	0.96	0.98	0.97	221
Potato_Early_blight	0.97	0.99	0.98	150
Potato_Late_blight	0.88	0.89	0.88	150
Potato_healthy	1.00	0.74	0.85	23
Tomato_Bacterial_spot	0.91	0.97	0.94	319
Tomato_Early_blight	0.92	0.67	0.77	150
Tomato_Late_blight	0.86	0.93	0.90	287
Tomato_Leaf_Mold	0.96	0.92	0.94	143
Tomato_Septoria_leaf_spot	0.97	0.94	0.95	266
Tomato_Spider_mites_Two_spotted_spider_mite	0.89	0.96	0.92	252
...				
accuracy			0.93	3096
macro avg	0.94	0.91	0.92	3096
weighted avg	0.93	0.93	0.93	3096

Fig. 6. Precision, recall, and F1-score.

timizer (lr=0.002). AdamW (3e-4), cosine annealing, label smoothing (0.05), and mixed-precision (AMP) are examples of ablations [20]. Reproducibility is ensured by artifacts (class_names.json, label encoder, configs); the best model (lowest val loss) is preserved as best_model.pth.

E. Integration and Deployment

AgriConnect incorporates Flask (inference), Express (validation, routing), React (upload, visualization), and MongoDB (users, predictions, marketplace listings) [21]. {class, confidence, top-k, advice, latency_ms} is returned by the API. MIME/type checks, size restrictions, JWT, and rate limiting are examples of security measures [22].

VI. EXPERIMENTAL SETUP

A. Hardware and Software

A GPU with CUDA support was used for training, and both CPU and GPU were used to evaluate inference. For this work NVIDIA DGXA100 supercomputer is used [23] Model size on disk, peak RAM, images/s (batch=1), and end-to-end latency from UI click to JSON answer are all reported.

B. Metrics

The performance measure for the work is [24] accuracy, Macro-Precision, Macro-Recall, Macro-F1, per-class F1, confusion matrix, robustness to perturbations, Expected Calibration Error (ECE), latency statistics (mean/p50/p95), and throughput (req/s) are all measured and the detailed report is mentioned in Fig. 6.

VII. RESULTS AND DISCUSSION

The proposed CNN attains a test accuracy of **93.06%**, a test loss of **0.2453**, and a macro-F1 score of **0.915** Learning curves indicate steady convergence without signs of overfitting, as shown in Fig. 7.

Comprehensive evaluations were conducted across multiple components including optimizer choices (SGD vs. AdamW), schedulers (none vs. cosine), loss functions (CE vs. label smoothing), augmentation strategies

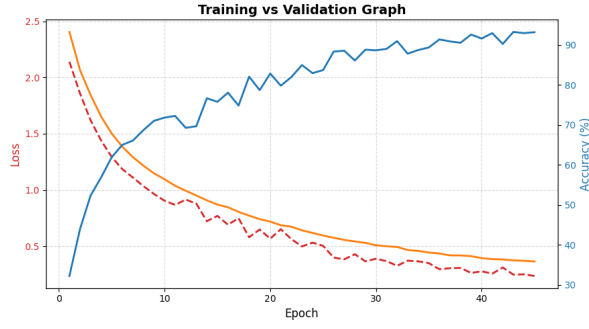


Fig. 7. Training and validation learning curves.

TABLE III
ABLATION SUMMARY (ILLUSTRATIVE).

Variant	Acc.	F1	Latency	Notes
Baseline (SGD, CE)	93.06	0.915	1.18s	—
+ AdamW (3e-4)	93.3	0.918	1.21s	Slight gain
+ Label Smoothing	93.2	0.919	1.18s	Calibrates conf.
+ Strong Augment	93.1	0.917	1.20s	Robustness ↑
w/o BN or w/o GAP	90–91	0.89–0.90	1.15s	Gen. drop

diction confidence closely aligned with actual accuracy. Entropy-based confidence estimation integrated into the user interface identifies low-confidence samples, prompting users to retake images or seek expert consultation. Calibration techniques such as label smoothing and temperature scaling further enhance prediction reliability and user trust.

The test accuracy of the proposed convolutional neural network is 93.06 %. , the test loss and macro-F1 score are 0.2453 and 0.915 respectively. With multi-worker Flask deployment, the system achieves a throughput of approximately 40 requests per second. The lightweight 11 MB model footprint makes the solution highly suitable for deployment in resource-constrained rural environments.

TABLE IV
COMPARATIVE ANALYSIS WITH BASELINE MODELS

Model	Accuracy	Parameters	Inference Time	F1-Score
Proposed CNN	93.06%	424K	45 ms	92%
ResNet-50	89.23%	23.5M	120 ms	88.75%
MobileNetV2	87.56%	2.2M	65 ms	86.91%
VGG-16	90.12%	134.3M	180 ms	89.45%
EfficientNet-B0	91.34%	4.0M	85 ms	90.87%

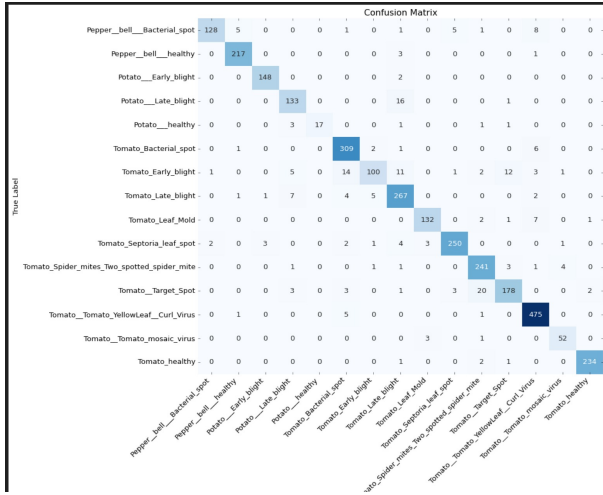


Fig. 8. Confusion matrix for the test dataset, where a darker diagonal reflects higher correct classification.

(baseline vs. jitter vs. random-resized-crop), regularization (varying dropout, with/without GAP and BN), and mixed-precision training (AMP). Table III summarizes the ablation findings.

Model robustness remains strong under challenging real-world distortions, achieving 87–92% accuracy in poor lighting, blur, and $\pm 15^\circ$ rotations. Challenging cases encompass overexposed. Classes, leaves, early-stage, soil-colored lesions, and classes of high visual similarity (e.g. nutrient deficiency vs. disease). Fig 8 shows the confusion matrix for the test dataset that reflects higher correct classification.

Expected Calibration Error (ECE) and reliability plots demonstrate that the model is well-calibrated, with pre-

In contrast to the traditional accuracy-based studies, the proposed ones are not accuracy-centric. A comparative analysis of proposed with the baseline models is contained in Table IV, which demonstrates the efficiency of the proposed approach. The system includes deployment factors such as pragmatic deployment. As latency, throughput, calibration behaviour and robustness analysis. The incorporation of an employment agency and the renting of agricultural machinery bridges the gap between diagnosis to actionable intervention. The architecture, including a modular MERN frontend, a Flask-based inference API, and potential Grad-CAM visualisations, multilingual UI, and support. On-device inference—scalable, supportable, optimized. In the real world of agricultural processes. This diagnosis-to-action. The time-to-intervention can be greatly minimised because of the pipeline. Especially important in the case of fast disease outbreaks.

VIII. CONCLUSION AND FUTURE WORK

AgriConnect, a small convolutional neural network-based system, is proposed by us that is designed to handle the seamless transformation of field image acquisition into accurate plant disease diagnosis as a part of an end-to-end MERN and Flask deployment setup. The proposed framework goes beyond the traditional studies, which are based on accuracy. It provides a feasible plan for scalable digital plant-health management due to the prioritization of deployability, interpretability, and actionable insight. The system has a test accuracy of 93.06 percent, a test loss of 0.2453, and a macro-F1 score of 0.915, has a lightweight model footprint of around 11MB, and an average end-to-end inference latency of 1.18 seconds,

showing that it is practicable in real-world agricultural applications. Its operational reliability under deployment constraints is further supported by robustness tests, calibration tests, and latency profiling tests. Regardless of such positive results, some shortcomings remain: the model depends on a high-quality image of leaves, it is hard to differentiate rare or similar-looking illnesses properly, and it requires a stable Internet connection to perform its inference in real time. Future studies will therefore focus on incorporating Grad-CAM explainability into the user interface to strengthen transparency and farmer trust, domain adaptation and fine-tuning methods to strengthen resilience to field variation, and supporting the on-device inference with the help of TensorFlow Lite and model quantization to reduce reliance on the network. Furthermore, the integration of drone-derived sensors and IoT will facilitate the multimodal, environment-conscious diagnosis, and privacy-preserving federated learning might facilitate decentralized model upkeep and safety. The extension of multilingual and voice-based interfaces is expected to be another factor to enhance more inclusive usage in rural areas with low levels of digital literacy.

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